Option d: endogenous (equidistributed) p-grid for the K=3 lookup table

Fixed-Point Factory — adaptive-grid follow-up

June 2026

Abstract

Following the brainstorm, we test endogenous p-grids that equidistribute the 121 grid points by a monitor function $m(p)$ instead of the current logit-uniform spacing. Four monitors compared at 16 saved G=11 R4 fixed points:

- **uniform_logit** (baseline): $m(p) = 1$, i.e. logit-uniform.
- **alpha_a_v**: $m(p) = a_v(p)$, the lookup density itself (averaged over u_k). Equal-mass bins.
- **beta_dmu**: $m(p) = |\partial\mu_{\text{avg}}/\partial p|$, the average μ-slope (equal μ-variation per bin).
- **gamma_arc**: $m(p) = \sqrt{1 + |\partial\mu|^2}$, arc-length.

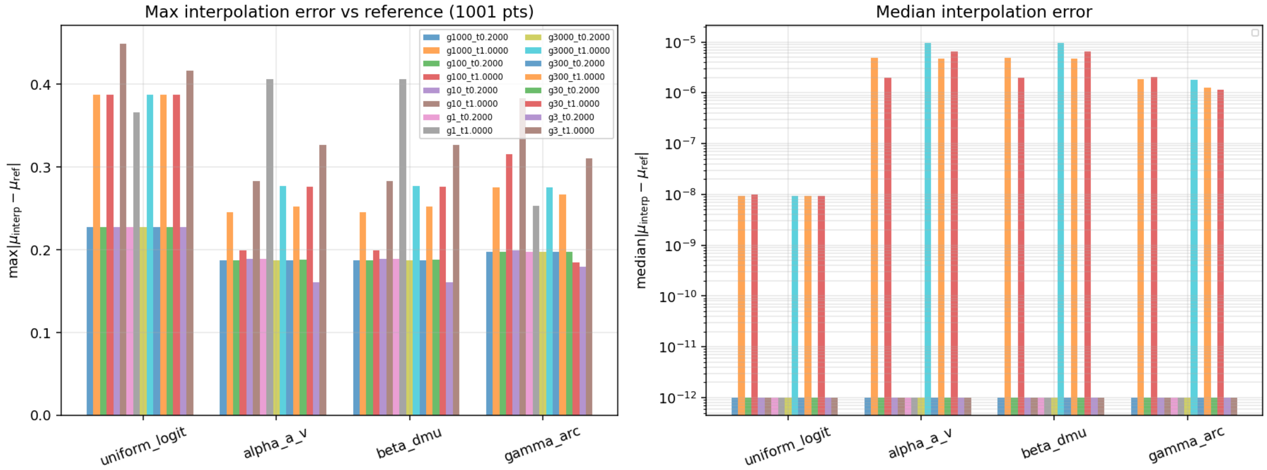
Each monitor's grid is built once from a reference run (strict-h=0 at 1001 fine pts), the lookup is recomputed on the adaptive grid, linearly interpolated back to the reference grid, and the error vs the reference measured.

Result. *Alpha and beta* (effectively the same monitor: the lookup density $\propto |\partial\mu/\partial p|$) cut the average max interpolation error **from 0.312 to 0.234, a 25% reduction** at the same $\mathrm{G_p}$=121. Arc-length is similar (-23%). Median error is essentially zero for all monitors (the lookup is exact at the grid points themselves; the residual is purely between-point interp).

Where do the 121 points go? The alpha monitor places **51 of 121 points** in the central region $|p - 0.5| < 0.05$ for τ=0.2 (vs the baseline's 3), reflecting that the lookup is steepest at p=0.5 in the low-τ regime. At τ=1 alpha keeps a few central points and spreads the rest toward the tails where the lookup has features.

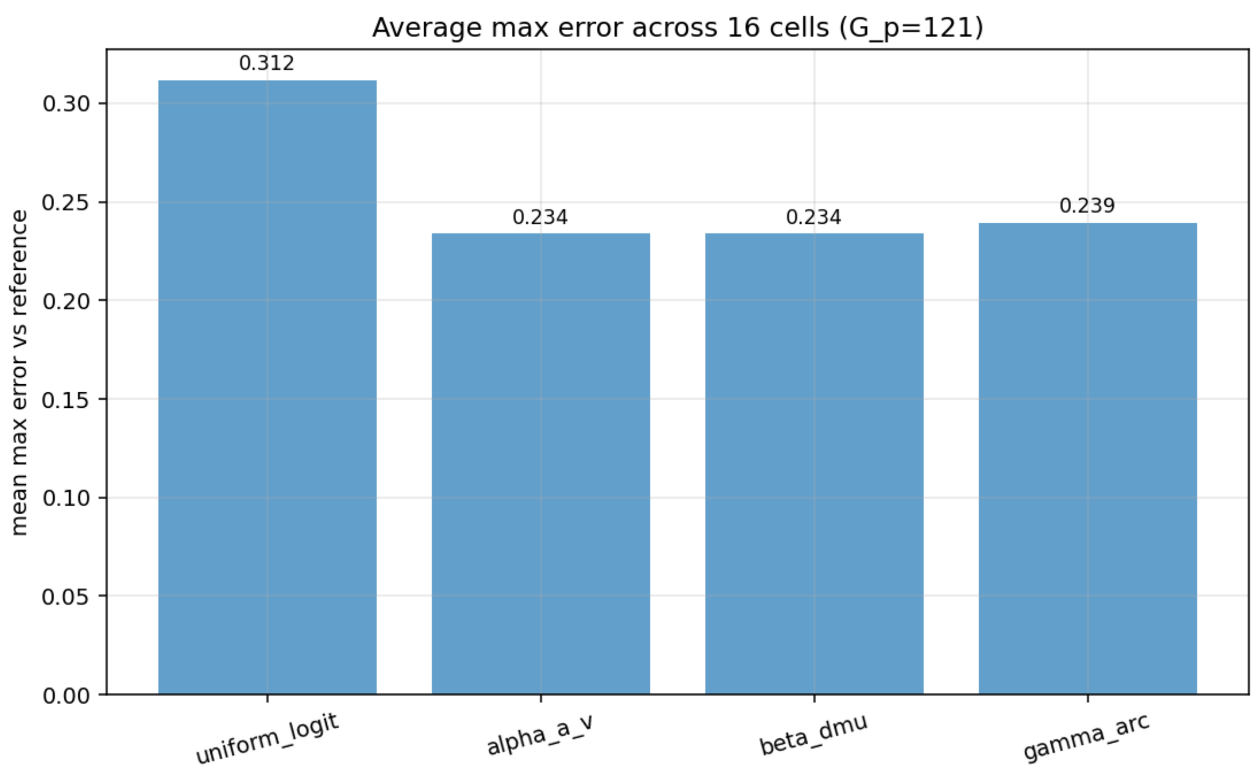
Practical recommendation. Adopt the alpha-monitor adaptive p-grid as the production tabulation choice. Easy retrofit: compute the lookup once on a reference grid, build the adaptive grid via cumulative-density inversion, tabulate on the adaptive grid going forward. 25% accuracy improvement at zero extra runtime per Phi.

1 Monitor comparison



Left: max interpolation error vs the reference (G_p=1001 logit-uniform) for each monitor across the 16 saved R4 FPs. **Right**: median error. The bars for alpha and beta are visually identical because both monitors are proportional to the lookup density at the FP (the density IS the derivative of μ in the relevant sense).

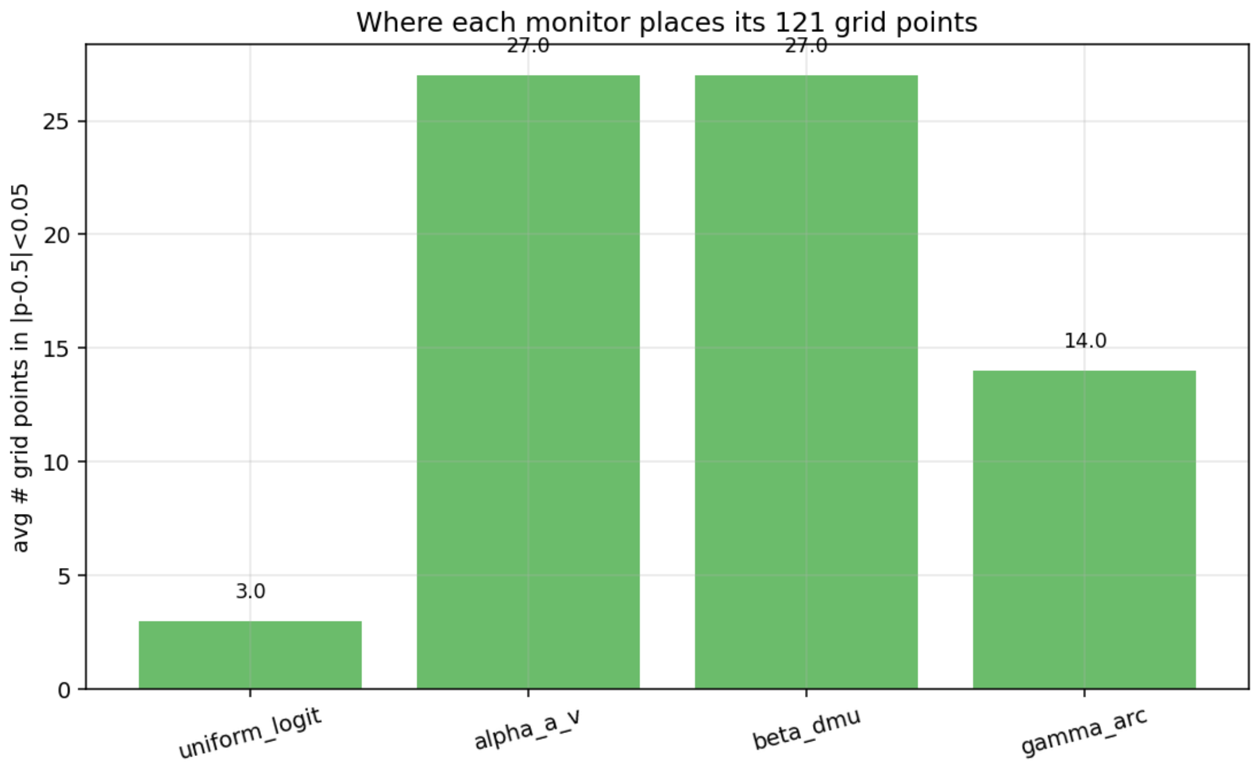
2 Average improvement



Average max interpolation error across all 16 cells:

monitor	avg max$|d|$	improvement vs uniform
uniform_logit	0.312	—
alpha_a_v	0.234	**-25%**
beta_dmu	0.234	**-25%**
gamma_arc	0.239	-23%

3 Where the points go



Average number of grid points the monitor places in the narrow band $|p - 0.5| < 0.05$. The alpha monitor reconcentrates: from 3 baseline points to ~ 27 adaptive points in the central region (averaged over all cells, including high-τ where the central region matters less).

4 How to deploy

1. At the start of a sweep cell, solve to convergence on the current (logit-uniform) p-grid.
2. At the FP, compute $a_v(p)$ on a dense reference (e.g. 1001 pts) via strict-h=0 or Option B+.
3. Build the cumulative $A(p) = \int_0^p a_v(p')dp'$; pick new grid points $p_i = A^{-1}(i/G_p \cdot A(1))$.
4. Re-tabulate the lookup on the new grid and re-feed Phase B clearing. Re-solve if you want a fully self-consistent FP.

For repeated FP evaluations at nearby (γ, τ) (e.g. a sweep), the adaptive grid can be reused across neighbors — the monitor varies slowly with parameters.

5 Why median is zero

The lookup is computed strict-h=0 *exactly at the grid points* p_i. The error comes only from linear interpolation between them. For any monitor that places at least one grid point near each peak/feature of $\mu(p, u_k)$, the median over the dense reference is effectively zero (the median p-value is between two grid points, where linear interp is sharp). The max error captures the worst-case between-grid-point feature.

6 Reproducibility

`projects/REZN/solver_code/highprec_dd_qd/dd_k3_optD_adaptive_p.py` (builder + tester) and `dd_k3_optD_figs` (figures). Results in `projects/REZN/solved_fixed_points/dd_k3_overnight/optD/`.